



Jeong-Yun and Christoph Keplinger demonstrate their ionic speaker

The devices apparently use the wireless signals as both a source of power and a medium for communication. Their usage could range from wearable computers to smart homes.

Harvard engineers develop stretchable speakers

A group of engineers from Harvard, led by Jeong-Yun and Christoph Keplinger, has developed a stretchable ionic speaker made of gel. The speaker is transparent and can play the entire audible sound spectrum, that is, 20 Hz to 20 kHz. The engineers have used a transparent piece of insulating and stretchable rubber along with a couple of layers of a hydrogel, which is an ionic conductor. This apparatus, in this case, has been attached to copper electrodes.

According to reports, this is the first device to prove that an ionic apparatus can be used in order to power conventional electronic devices. The engineers believe that the device could be attached to a TV to eliminate the need for speakers. The stretchable speaker is also almost invisible. The idea of ionic conductors has many applications in the real world. In fact, it can be used right from the most basic electronics to the complete artificial organs, which would save many lives.

Indian student designs UAV

Subhanjan Saha, a 21-year-old computer-science student from Mahindra Chandra College, has developed a 3*1m completely unmanned aerial vehicle (UAV). His design is based on basic materials but is highly capable of camera surveillance attack and defense, weather forecasting and much more.

He has already written to the Defense Research and Development Organisation (DRDO) and the Hindustan Aeronautics Limited (HAL), and is waiting for the authorities to fund the upgradation of the drone.

The drone is currently equipped with a pair of 180 degrees high-definition cameras (each in the front and back) and, once airborne, can scan a 10-kilometre area. The vehicle is totally remote controlled. Saha explains that his drone runs on re-

chargeable batteries and is completely eco-friendly. It can be airborne for an average stretch of 60 minutes and, if funding is established, the design can be further improved to make it last longer.

Ultra-fast camera which works like human retina

Photos and videos taken by modern DSLR cameras are nothing close to light-weight files. They quickly add up to gigabytes of data, which makes it difficult for the researchers. However, engineers at the inILabs, a Swiss company, have come out with a technology that borrows the image capturing mechanics from the way a human retina functions. Aptly called the dynamic vision sensor (DVS), the resulting technology is the prominent aspect of this super-powerful and ultra-fast camera.

The human retina works not by actually capturing/recording all the things that come to our view but by spotting the change in movements. This way the neurons can get rid of tons of unnecessary data from the surrounding, making processing sharper and fast. The DVS technology works the same way and records only motion of the subjects. The new camera is able to record hours of non-stop video consuming a very little amount of power and using just a few megabytes for storage of the same.

Bracelet that confirms identity from heartbeats

A new device has been developed by a company called Bionym, which identifies users by reading their heartbeats.



Nyimi bracelet, which identifies users through their heartbeat